

# omicstra TNBC-92

## tnbc-92 · task-conditional routing matrix (v2 - v3 gpath2vec retrain)

which alignment to use for which question, with v3 evidence. routing-rule **shape** unchanged from v1; only the cited numbers were updated against the v3 retrain.

**v1 baseline** for head-to-head: [../v1/tnbc92\\_routing\\_matrix.md](#).

### routing rule (v3 evidence)

| if the question is ...                                 | route to ...                            | v3 evidence                                                                                                   | construct-validity guard                                                                                                 |
|--------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| cross-modal retrieval (find matched ST niche from H&E) | <b>R4_v3</b>                            | H1 AUC 0.859, CKA 0.631, alignment gap 0.233. rank-matched control still rejects collapse-driven explanation. | cross-subarray patient-held-out is the only honest retrieval eval (within-subarray = patient leakage).                   |
| group niches by recurring tumor-microenvironment state | <b>R5_v3 / R6_v3 / R1_v3</b> (3-way tie | H2 KMeans on mc_megacluster (Wang per-spot 14-class NMF, niche-level): R5 ARI                                 | use mc_megacluster not archetype (archetype is patient-level pseudobulk, $NMI(\text{archetype}, \text{patient}) \approx$ |

| if the question is ...                                             | route to ...                                                                            | v3 evidence                                                                                                                                                                                                                                                                               | construct-validity guard                                                                                                                                                                    |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| amplify biology, suppress patient identity                         | within seed noise)<br><br><b>R6_v3 / R5_v3 (best ratio) or R4_v3 (lowest patient z)</b> | 0.246, R6<br>0.244, R1<br>0.234.<br><br>v3 bio/patient (TIME z / patient z, z_he): R6 0.547, R5 0.497, R1 0.461, R4 0.376; B1 0.071 below raw_he floor 0.137. R4 patient z 21.5 (lowest in grid).<br><br>H3 gpath2vec arm option A (cross-patient held-out, BH-FDR<0.05): R4 4/4 testable | 0.89 - any “win” reads patient identity).<br><br>pair every ARI / silhouette claim with a bio-z vs patient-z permutation z-test (projects/tnbc-92/evaluation_question_audit.md finding #3). |
| find pathway-similar niches across new patients (proposal-literal) | <b>R4_v3</b>                                                                            | pathways significant, z_A 11.2-25.4 (Immune 25.4, ECM 16.0, CC 16.5, PCD 11.2). late-fusion contrastive (R1/R3 1/4, R2/R6 0/4, R5 2/4) fail or trail.                                                                                                                                     | report z_he only (z_st / z_mean circular with v3 gpath2vec on ST input). TGF- $\beta$ set_size=2 underpowered, excluded from FDR family.                                                    |
| detect fine-grained sub-pathway biology, esp. dense linear ECM     | <b>B1_v3 / B2_v3</b> (classical) or <b>R4_v3</b> (competitive)                          | gpath2vec arm: B2/B1 4/4 testable, ECM z_A 9.2-10.4. R4 wins overall but classical is                                                                                                                                                                                                     | classical baselines pass H3 cross-patient on the linear ECM signal; use them as the right tool for that question even though they fail H1.                                                  |

| if the question is ...                                      | route to ...                                                                       | v3 evidence                                                                                                                                                                                                                                                   | construct-validity guard                                                                                                                                                                                                                                          |
|-------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FTU-coherence - TLS compartment (categorical, H2 Part B.1)  | <b>R6_v3 / R2_v3 / R1_v3 / R5_v3 / B2_v3</b>                                       | comparable on ECM.<br>v3 Welch z: TLS-vs-TIL passes in R6 (+5.6/+5.9), R2 (+5.4/+5.4), R1 (+2.2/+4.1), R5 (+2.4/+3.2), B2 (+3.7/+2.8). Tumor/ Necrosis invert (annotation-granularity confound). R3 uniquely passes Necrosis (+5.8/+9.1). R4 fails all (0/6). | late-fusion contrastive optimizes within-compartment matched-pair cosine by construction -> wins on the well-annotated FTU (TLS).                                                                                                                                 |
| FTU detection - continuous TLS gene signature (H2 Part B.2) | <b>B4_v3 (random-init) / B1_v3 / R4_v3 / B2_v3</b> - avoid late-fusion contrastive | v3 z_he TLS-signature CCA (cross-patient, BH-FDR<0.05): B4 z_TLS 25.5, B1 16.6, R4 11.4, B2/B3 11.3, R5 2.4 (marginal); R1/ R2/R3/R6 fail.                                                                                                                    | TLS is directly H&E-morphology-decodable -> any method preserving Virchow2 features works (even untrained). late-fusion contrastive <i>degrades</i> z_he linear-decodability - a documented cost of the contrastive objective on single-modality-encoded signals. |
| avoid for cohort-level biology interpretation               | <b>B1_v3 (CCA)</b>                                                                 | v3 H1 AUC 0.544 (weakest contrastive-family-comparable). CCA amplifies patient identity by construction:                                                                                                                                                      | pair every ARI report with a bio-z vs patient-z ratio; ratio below the raw-modality floor means patient identity dominates - anti-helpful for cohort use.                                                                                                         |

| if the question is ...                                                | route to ...                                                              | v3 evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | construct-validity guard                                                                                                      |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| sanity floor /<br>isolate alignment<br>from architecture              | <b>B3_v3</b><br>(unaligned<br>PCA) + <b>B4_v3</b><br>(random-init<br>MLP) | v3 patient z<br>113.0 (~3×),<br>bio/patient<br>ratio 0.071<br>below raw_he<br>floor 0.137.<br><br>v3 B3 AUC<br>0.441 sub-<br>chance, gap<br>-0.053 (correct<br>sanity<br>behavior). v3<br>B4 AUC 0.496 ≈<br>chance<br>(architecture<br>alone<br>contributes<br>none of H1<br>gain).<br>v3 AUC 0.706.<br>B2 and B3 are<br>rotation-<br>equivalent on<br>H2 (within-<br>space metrics)<br>and yield<br>identical<br>specificity<br>matrices; they<br>diverge on H1<br>(cross-modal<br>cosines<br>depend on<br>rotation). | B3 / B4 are required falsifiers<br>- any aligned method must<br>beat both decisively.                                         |
| classical<br>alignment family<br>with rotation-<br>equivalence sanity | <b>B2_v3</b><br>(Procrustes)                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | use this divergence to verify<br>the test scoring is correct - if<br>B2 = B3 on a cross-modal<br>metric, something is broken. |

## v3 vs v1 routing changes

the core routing rule **shape is unchanged from v1** - R4 for retrieval + cross-patient pathway transfer, late-fusion (R1/R5/R6) for niche clustering, B1 anti-helpful for cohort biology. numbers improved or held (R4 H1 AUC 0.851 -> 0.859; R4 H3 gpath2vec z\_A higher than v1's AUCell arm; bio/patient ratios reproduce v1). **two additions in v3:**

- **R5 (AnInfoNCE + late fusion)** joins the niche-clustering tie (R5/R6/R1) and the bio/patient leaders; it does not displace any method.
  - **TLS FTU detection splits into two routes (B.1 vs B.2)** - the categorical compartment test (B.1) routes to late-fusion contrastive (they optimize within-compartment cosine); the continuous gene-signature test (B.2) routes to morphology-preserving methods (random-init, classical, R4) and **explicitly avoids late-fusion contrastive**, which degrades H&E linear-decodability. this is the one place a method's routing flips between two operationalizations of the same FTU - a new v3 finding.
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## v3-specific guards to encode in the MCP agent

these are construct-validity guards lifted directly from the v3 evidence and the audit doc - the MCP agent must enforce them (see docs/mcp\_agent\_design.md for the design spec):

- **archetype-ARI is patient-leakage diagnostic, not biology** - never report it as a biology coherence claim without paired mc\_megacluster ARI + patient-z diagnostic.
- **z\_st and z\_mean are circular** on H3 metrics when gpath2vec is on the ST input (R1/R2/R3/R4, B1/B2/B3 on v3). H3 z\_he is the only clean view.
- **mc\_megacluster is also partially circular on z\_st** (ST input encodes the data the NMF clustered). z\_he mc test is the citable cross-modal version.
- **R4 KMeans low MC ARI is metric-induced** (manifold geometry vs KMeans globular assumption), not biology absence. v3 patient-stratified linear probe confirms: R4 0.240 at parity with R1 0.282 / raw\_he 0.270 despite KMeans ARI 0.101.
- **gpath2vec\_niche on ST input is sha-locked to v3 build**  
13985cbd... per data/embeddings/niches\_v3/manifest.json. do not silently swap.

- **TGF- $\beta$  R-HSA-170834 is structurally underpowered** on the v3 build (set\_size=2) - exclude from H3 FDR family, report separately.
  - **R6 is the no-gpath2vec ablation** - if R6 H3 z\_A >> 0, the gpath2vec ST anchor is not load-bearing; v3 confirms R6 0/4 testable, so the anchor IS load-bearing for cross-patient pathway transfer.
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## pointers

| asset                                                  | path                                                                                                      |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| v1 routing matrix                                      | <a href="#">../v1/tnbc92_routing_matrix.md</a>                                                            |
| v3 results summary                                     | <a href="#">tnbc92_results_summary.md</a>                                                                 |
| MCP / LangGraph agent design<br>(encodes these guards) | <a href="#">docs/mcp_agent_design.md</a>                                                                  |
| evaluation-question audit<br>(Cameron's questions)     | <a href="#">projects/tnbc-92/<br/>evaluation_question_audit.md</a>                                        |
| proposal deviations log                                | <a href="#">projects/tnbc-92/<br/>proposal_deviations.md</a>                                              |
| v3 retrain plan + memory                               | <a href="#">projects/tnbc-92/<br/>v3_phase2_plan.md,</a><br><a href="#">[[project_v3_retrain_result]]</a> |